

MB-TaylorFormer: Multi-branch Efficient Transformer Expanded by Taylor Formula for Image Dehazing

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Abstract

In recent years, Transformer networks are beginning to replace pure convolutional neural networks (CNNs) in the field of computer vision due to their global receptive field and adaptability to input. However, the quadratic computational complexity of softmax-attention limits the wide application in image dehazing task, especially for high-resolution images. To address this issue, we propose a new Transformer variant, which applies the Taylor expansion to approximate the softmax-attention and achieves linear computational complexity. A multi-scale attention refinement module is proposed as a complement to correct the error of the Taylor expansion. Furthermore, we introduce a multi-branch architecture with multi-scale patch embedding to the proposed Transformer, which embeds features by overlapping deformable convolution of different scales. The design of multi-scale patch embedding is based on three key ideas: 1) various sizes of the receptive field; 2) multi-level semantic information; 3) flexible shapes of the receptive field. Our model, named Multi-branch Transformer expanded by Taylor formula (MB-TaylorFormer), can embed coarse to fine features more flexibly at the patch embedding stage and capture long-distance pixel interactions with limited computational cost. Experimental results on several dehazing benchmarks show that MB-TaylorFormer achieves state-of-the-art (SOTA) performance with a light computational burden. The source code and pre-trained models are available at <https://github.com/FVL2020/ICCV-2023-MB-TaylorFormer>.

1. Introduction

Single image dehazing is an image restoration task which aims to estimate latent haze-free images from hazy images. Starting with early CNN-based approaches [49, 5]

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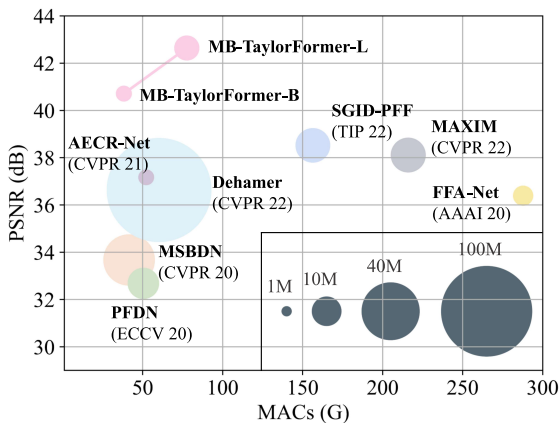


Figure 1: **Improvement of MB-TaylorFormer over the SOTA approaches.** The circle size is proportional to the number of model parameters. All models are trained on SOTS-Indoor [32].

and their revolutionary performance in dehazing, haze removal gradually shifts from prior based strategies [89, 20] to deep learning-based methods. In the past decade, deep dehazing networks achieve significant performance improvement due to advanced architectures like multi-scale information fusion [49, 47, 37], sophisticated variants of convolution [74, 36], and attention mechanisms [45, 86, 21].

Recently, Transformer has been popularly employed in various computer vision tasks and subsequently contributes greatly to the progress of high-level vision tasks [39, 71, 61, 6]. In the field of image dehazing, there are several challenges with the direct application of Transformer: 1) the computational complexity of Transformer is quadratic with the resolution of feature map, which makes it poorly suited to the pixel-to-pixel task of dehazing. Although some works apply self-attention in small spatial windows [35, 73] to relieve this problem, the receptive field of Transformer is restricted; 2) the basic elements of visual Transformer usually have more flexible scales [39]. However, existing

MB-TaylorFormer: 基于泰勒公式展开的多分支高效Transformer用于图像去雾

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摘要

近年来, *Transformer*网络因其全局感受野和对输入的适应性, 开始在计算机视觉领域取代纯卷积神经网络 (*CNN*)。然而, *softmax*注意力的二次计算复杂度限制了其在图像去雾任务中的广泛应用, 尤其是对于高分辨率图像。为了解决这一问题, 我们提出了一种新的 *Transformer*变体, 它应用泰勒展开来近似*softmax*注意力, 并实现了线性计算复杂度。我们提出了一个多尺度注意力细化模块作为补充, 以修正泰勒展开的误差。此外, 我们在所提出的*Transformer*中引入了具有多尺度块嵌入的多分支架构, 该架构通过不同尺度的重叠可变形卷积来嵌入特征。多尺度块嵌入的设计基于三个关键思想: 1) 感受野的各种尺寸; 2) 多层次语义信息; 3) 感受野的灵活形状。我们的模型名为*MB-TaylorFormer* (*Taylor formula expanded Multi-branch Transformer*), 可以在块嵌入阶段更灵活地嵌入从粗到细的特征, 并以有限的计算成本捕获长距离像素交互。在几个去雾基准数据集上的实验结果表明, *MB-TaylorFormer*以较轻的计算负担实现了最先进 (*SOTA*) 的性能。源代码和预训练模型可在 <https://github.com/FVL2020/ICCV-2023-MB-TaylorFormer>。

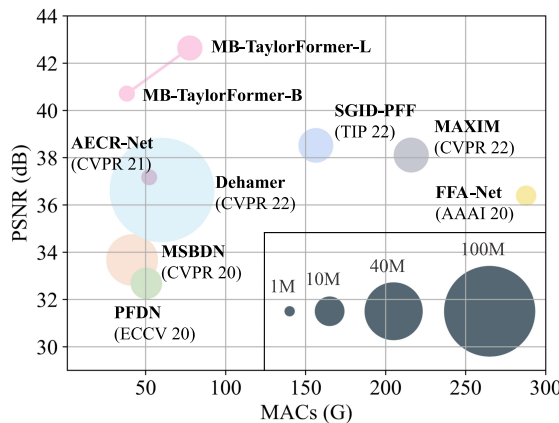


图 1: **MB-TaylorFormer 相较于 SOTA 方法的改进。** 圆圈大小与模型参数量成正比。所有模型均在 SOTS-Indoor [32] 上训练。

及其在去雾中的革命性表现开始, 去雾逐渐从基于先验的策略 [89, 20]转向基于深度学习的方法。在过去十年中, 深度去雾网络取得了显著的性能提升, 这归功于先进的架构, 如多尺度信息融合 [49, 47, 37], 复杂的卷积变体 [74, 36], 和注意力机制 [45, 86, 21]。

近年来, Transformer已被广泛应用于各种计算机视觉任务, 并随后极大地推动了高层视觉任务的进展 [39, 71, 61, 6]。在图像去雾领域, 直接应用 Transformer面临若干挑战: 1) Transformer的计算复杂度随特征图分辨率呈二次方增长, 这使其不适合去雾这类像素到像素任务。尽管一些工作在小空间窗口内应用自注意力 [35, 73]以缓解这一问题, 但Transformer的感受野受到了限制; 2) 视觉Transformer的基本元素通常具有更灵活的尺度 [39]。然而, 现有的

1. 引言

单幅图像去雾是一项图像复原任务, 旨在从有雾图像中估计潜在的无雾图像。从早期基于卷积神经网络的方法 [49, 5]

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visual Transformer networks [78, 73] generally generate fixed-scale tokens by fixed convolution kernels. Thus, there is still room for improvement via introducing flexible patch embedding Transformer to the dehazing task.

To address the first challenge, we propose a Transformer variant expanded by Taylor formula (TaylorFormer), which applies self-attention on the entire feature map across spatial dimension and maintains linear computational complexity. Specifically, we calculate the weights of self-attention by performing a Taylor expansion on softmax, and then reduce the computational complexity of self-attention from $\mathcal{O}(n^2)$ to $\mathcal{O}(n)$ by applying the associative law of matrix multiplication. This strategy brings three advantages: 1) it retains the ability of Transformer to model long-distance dependencies among data, and avoids reducing the receptive field caused by window splitting [58]; 2) it provides a stronger value approximation than methods using a kernel-based formulation of self-attention [28], and is similar to the vanilla Transformer [64]; 3) it makes the Transformer concerned with pixel-level interactions rather than channel-level interactions [78], which allows for more fine-grained processing of features.

Considering the error caused by ignoring Peano’s form of remainder [43] in the Taylor formula, we introduce a multi-scale attention refinement (MSAR) module to refine TaylorFormer. We exploit the local correlation within image by convolving the local information of queries and keys to output a feature map with scaling factors. The number of feature map channels is equal to the number of heads in multi-head self-attention (MSA), so each head has a corresponding scaling factor. Our experiments show that the proposed MSAR module effectively improves model performance with tiny computational burden (see Table 3).

To tackle the second challenge, inspired by the success of inception modules [60, 59] and deformable convolutions [13] in CNN-based dehazing networks [18, 65, 44, 74], we propose a multi-branch encoder-decoder backbone for TaylorFormer, termed ad MB-TaylorForemr, based on multi-scale patch embedding. The multi-scale patch embedding has various sizes of receptive field, multi-level semantic information, and flexible shape of receptive field. Considering that the generation of each token should follow the local relevance prior, we truncate the offsets of the deformable convolutions. We reduce computational complexity and the number of parameters by the depthwise separable method. The tokens from different scales are then fed independently into TaylorFormer and finally fused. The multi-scale patch embedding module is capable of generating tokens with different scales and dimensions, and the multi-branch structure is capable of processing them simultaneously to capture more powerful features.

To summarize, our main contributions are as follows: (1) We propose a new variant of linearized Transformer based

on Taylor expansion to model long-distance interactions between pixels without window splitting. An MSAR module is introduced to further correct errors in the self-attention of TaylorFormer; (2) We design a multi-branch architecture with multi-scale patch embedding. Among it, multiple field sizes, flexible shape of receptive field, and multi-level semantic information can help simultaneously generate tokens with multi-scales and capture more powerful features; (3) Experimental results on public synthetic and real dehazing datasets show that the proposed MB-TaylorForme achieves state-of-the-art (SOTA) performance with few parameters and MACs.

2. Related Works

CNN-based Image Dehazing. With the development of CNN, significant progress has been achieved in image restoration tasks [70, 85, 81, 84, 82, 67, 66], including image dehazing [5, 45, 19, 4, 88, 69]. One way is based on the atmospheric scattering model, such as DehazeNet [5], DCPDN [80], and Aod-Net [31]. However, the performance is hardly satisfactory when the atmospheric scattering model fails. The other way is based on the idea of image conversion, which is not dependent on the atmospheric scattering model. These kinds of methods predict haze-free images in an end-to-end manner, such as EPDN [47], FFA-Net [45] and AECRNet [74]. Nevertheless, it is difficult for CNN-based models to learn long-range pixel dependencies. **Efficient Self-attention.** The computational complexity of the Transformer grows quadratically with the increasing spatial resolution of feature map, which makes it very demanding on computational resources. Some works reduce the computational burden by sliding window [48, 24] or shifted window [39, 35, 73, 58] based self attention. However, this design limits the ability of Transformer to model long-distance dependencies in the data. MaxViT [63] compensates for the decrease in receptive field with Grid attention. However, Grid attention is not strictly linear in computational complexity and is still quadratic on high-resolution images. Another approach is to modify the softmax-attention of the vanilla Transformer. Restormer [78] applies self-attention between channels, and ignores global relationships between pixels. Performer [12] achieves linear complexity by random projection. However, the queries, keys, and values require a large size, which results in increasing computation cost. Poly-nl [3] bridges the connection between attention and high-order polynomials. However, this has not been explored in a self-attention structure. [28, 52, 46, 75, 57] decompose softmax by kernel functions and use the associative law of matrix multiplication to achieve linear complexity. These methods are functional approximations, e.g., each element in the attention map is positive [46]. However, value approximations are not considered. Our method does not use the kernel func-

视觉Transformer网络 [78, 73]通常通过固定的卷积核生成固定尺度的令牌。因此，通过将灵活的块嵌入Transformer引入去雾任务，仍有改进空间。

为了解决第一个挑战，我们提出了一种由泰勒公式扩展而来的Transformer变体（TaylorFormer），它在整个特征图上沿空间维度应用自注意力，并保持线性计算复杂度。具体来说，我们通过对softmax执行泰勒展开来计算自注意力的权重，然后通过应用矩阵乘法结合律，将自注意力的计算复杂度从 $\mathcal{O}(n^2)$ 降低到 $\mathcal{O}(n)$ 。这一策略带来了三个优势：1) 它保留了Transformer对数据间长距离依赖进行建模的能力，并避免了由窗口划分 [58]引起的感受野缩小；2) 它提供了比使用基于核的自注意力公式 [28], 更强的值近似，并且与原始Transformer [64]相似；3) 它使Transformer关注像素级交互而不是通道级交互 [78], 从而允许对特征进行更细粒度的处理。

考虑到在泰勒公式中忽略佩亚诺余项 [43] 所带来的误差，我们引入了一个多尺度注意力细化（MSAR）模块来细化TaylorFormer。我们通过对查询和键的局部信息进行卷积来利用图像内的局部相关性，从而输出一个带有缩放因子的特征图。特征图通道数等于多头自注意力（MSA）中的头数，因此每个头都有一个对应的缩放因子。我们的实验表明，所提出的MSAR模块以极小的计算负担有效地提高了模型性能（见表3）。

为了解决第二个挑战，受Inception模块 [60, 59] 和可变形卷积在基于CNN的去雾网络中成功应用的启发，[13] [18, 65, 44,74], 我们为TaylorFormer提出了一种基于多尺度块嵌入的多分支编码器-解码器骨干网络，称为MB-TaylorForemr。多尺度块嵌入具有不同大小的感受野、多层次语义信息以及灵活的感受野形状。考虑到每个令牌的生成应遵循局部相关性先验，我们对可变形卷积的偏移量进行了截断。我们通过深度可分离方法来降低计算复杂度和参数量。来自不同尺度的令牌随后被独立地馈入TaylorFormer，并最终进行融合。多尺度块嵌入模块能够生成具有不同尺度和维度的令牌，而多分支结构能够同时处理这些令牌以捕获更强大的特征。

总结而言，我们的主要贡献如下：（1）我们提出了一种基于泰勒展开的线性化Transformer新变体，用于在不进行窗口划分的情况下建模像素之间的长距离交互。引入了一个MSAR模块以进一步校正TaylorFormer自注意力中的误差；

（2）我们设计了一种具有多尺度块嵌入的多分支架构。其中，多样的感受野大小、灵活的感受野形状以及多层次语义信息有助于同时生成多尺度token并捕获更强大的特征；（3）在公开合成和真实去雾数据集上的实验结果表明，所提出的MB-TaylorFormer在参数量和乘法运算次数都很少的情况下，实现了最先进（SOTA）的性能。

2. 相关工作

基于CNN的图像去雾。随着CNN的发展，图像复原任务取得了显著进展 [70, 85, 81, 84, 82, 67, 66], , 包括图像去雾 [5, 45, 19, 4, 88, 69]。一种方法基于大气散射模型，例如DehazeNet [5]、DCPDN [80], 和AOD-Net [31]。然而，当大气散射模型失效时，其性能难以令人满意。另一种方法基于图像转换的思想，不依赖于大气散射模型。这类方法以端到端的方式预测无雾图像，例如EPDN [47]、FFA-Net [45] 和AECRNet [74]。然而，基于CNN的模型难以学习长距离像素依赖。**高效自注意力。**Transformer的计算复杂度随特征图空间分辨率的增加呈二次增长，这对计算资源要求很高。一些工作通过滑动窗口 [48, 24] 或移位窗口 [39, 35, 73, 58] 的自注意力来降低计算负担。然而，这种设计限制了Transformer建模数据中长距离依赖的能力。MaxViT [63] 通过网格注意力补偿感受野的减小。然而，网格注意力在计算复杂度上并非严格线性，在高分辨率图像上仍是二次的。另一种方法是修改原始Transformer的softmax注意力。Restormer [78] 在通道间应用自注意力，并忽略像素间的全局关系。Performer [12] 通过随机投影实现线性复杂度。然而，查询、键和值需要较大的尺寸，这导致计算成本增加。Poly-NL [3]建立了注意力与高阶多项式之间的联系。然而，这尚未在自注意力结构中得到探索。[28, 52, 46, 75, 57]通过核函数分解softmax，并利用矩阵乘法的结合律实现线性复杂度。这些方法是函数近似，例如注意力图中的每个元素都是正的 [46]。然而，值近似未被考虑。我们的方法不使用核函

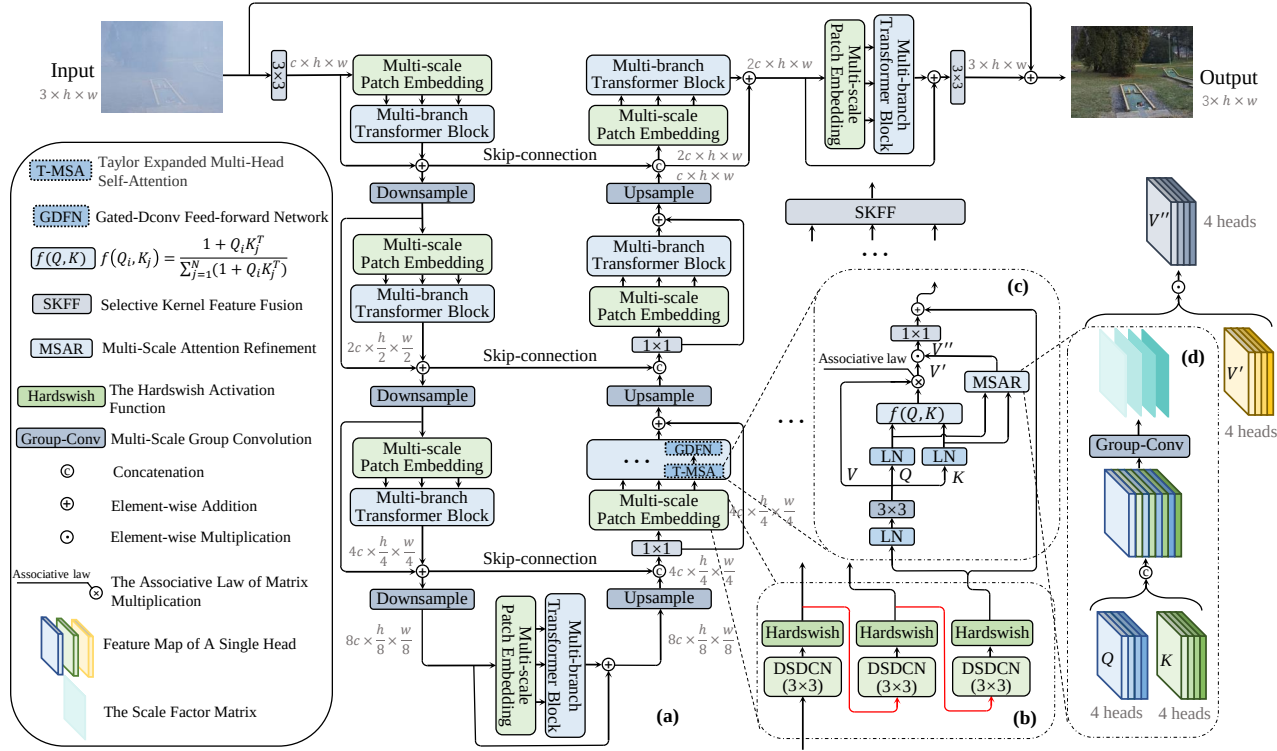


Figure 2: **Architecture of MB-TaylorFormer.** (a) MB-TaylorFormer consists of a multi-branch hierarchical design based on multi-scale patch embedding. (b) Multi-scale patch embedding embeds coarse-to-fine patches. (c) TaylorFormer with linear computational complexity. (d) MSAR module compensates for errors in Taylor expansion.

tion but performs the Taylor expansion on softmax directly, which guarantees a functional and numerical approximation to softmax.

Multi-scale Transformer Networks. In the field of high level vision, [41] is a simple pyramid structure. IFormer [55] applies inception structures to mix high and low frequency information. However, it does not utilize different patch sizes. CrossViT [7] and MPViT [30] process multi-scale patches via multiple branches to obtain multi-scale receptive fields. However, the receptive field shape is not flexible. In the field of low level vision, MSP-Former [76] uses multi-scale projections to help Transformer to represent complex degraded environment. Gique [54] processes feature maps of different sizes via multi-branch. [87] represents different features related to the task utilizing multiple sub-network. The recent Transformer networks for recovery tasks [78, 73, 58, 26, 41] build a simple U-net network or with single-scale patches. However, these works hardly further explore multi-scale patches and multi-branch architectures. Although [29] uses deformable convolution in self-attention, the number of sampling points of the convolution kernel is fixed. While our multi-scale deformable convolution not only has flexible sampling points, but also provides multi-level semantic in-

formation.

3. MB-TaylorFormer

We aim to build an efficient and lightweight Transformer-based dehazing network. To reduce the computational complexity, we apply Taylor expansion of softmax-attention to satisfy the associative law and adopt a U-net structure similar to the Restormer [78]. To compensate for the effects of Taylor expansion errors, we propose an MSAR module. In the following parts, we first describe the overall architecture of MB-TaylorFormer (Fig. 2a). Then we introduce three core modules: multi-scale patch embedding (Fig. 2b), Taylor expanded self-attention (Fig. 2c) and MSAR module (Fig. 2d).

3.1. Multi-branch Backbone

Given a hazy image $I \in \mathbb{R}^{3 \times h \times w}$, we apply convolution for shallow feature extraction to generate $F_0 \in \mathbb{R}^{c \times h \times w}$. Subsequently, we employ a four-stage encoder-decoder network for deep feature extraction. Each stage has a residual block containing a multi-scale patch embedding and a multi-branch Transformer block. We use multi-scale

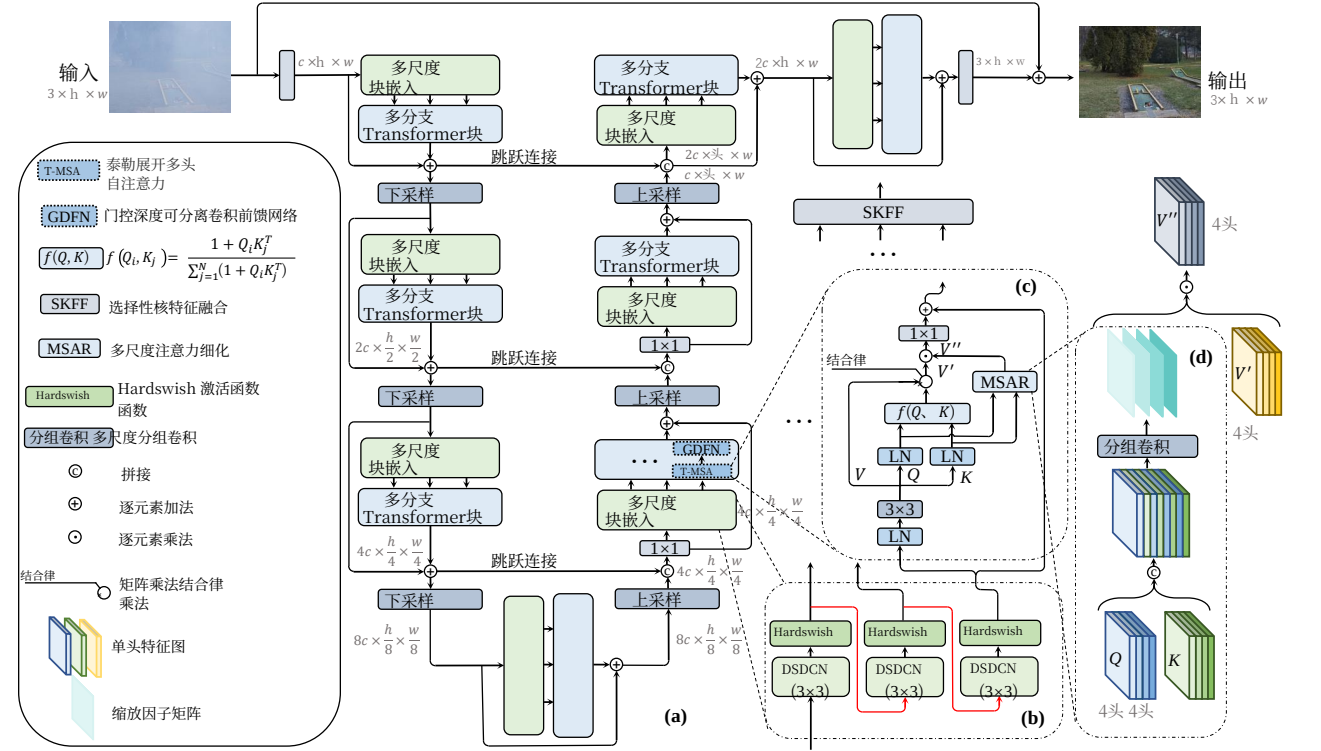


图2: **MB-TaylorFormer的架构。** (a) MB-TaylorFormer由基于多尺度块嵌入的多分支分层设计组成。 (b) 多尺度块嵌入嵌入从粗到细的图像块。 (c) 具有线性计算复杂度的TaylorFormer。 (d) MSAR模块补偿泰勒展开中的误差。

数，而是直接对softmax进行泰勒展开，这保证了对于softmax的函数近似和数值近似。

多尺度Transformer网络。在高层视觉领域，[41]是一种简单的金字塔结构。IFormer [55]应用Inception结构来混合高频和低频信息。然而，它并未利用不同的图像块尺寸。CrossViT [7]和MPViT [30]通过多个分支处理多尺度图像块以获得多尺度感受野。然而，感受野的形状并不灵活。在低层视觉领域，MSP-Former [76]利用多尺度投影帮助Transformer表示复杂的退化环境。Gique [54]通过多分支处理不同尺寸的特征图。[87]利用多个子网络表示与任务相关的不同特征。近年来用于恢复任务的Transformer网络[78, 73, 58, 26, 41]构建简单的U-net网络或使用单尺度图像块。然而，这些工作几乎没有进一步探索多尺度图像块和多分支架构。尽管[29]在自注意力中使用了可变形卷积，但卷积核的采样点数量是固定的。而我们的多尺度可变形卷积不仅具有灵活的采样点，还能提供多层级语义信

息。

3. MB-TaylorFormer

我们旨在构建一个高效且轻量级的基于Transformer的去雾网络。为了降低计算复杂度，我们对softmax注意力应用泰勒展开以满足结合律，并采用类似于Restormer的U-net结构[78]。为了补偿泰勒展开误差的影响，我们提出了一个MSAR模块。在接下来的部分中，我们首先描述MB-TaylorFormer的整体架构（图2a）。然后我们介绍三个核心模块：多尺度块嵌入（图2b）、泰勒展开自注意力（图2c）和MSAR模块（图2d）。

3.1. 多分支主干网络

给定一幅有雾图像 $I \in \mathbb{R}^{3 \times h \times w}$ ，我们使用卷积进行浅层特征提取，生成 $F_0 \in \mathbb{R}^{c \times h \times w}$ 。随后，我们采用四阶段编码器-解码器网络进行深层特征提取。每个阶段都有一个残差块，其中包含多尺度块嵌入和一个多分支Transformer块。我们使用多尺度

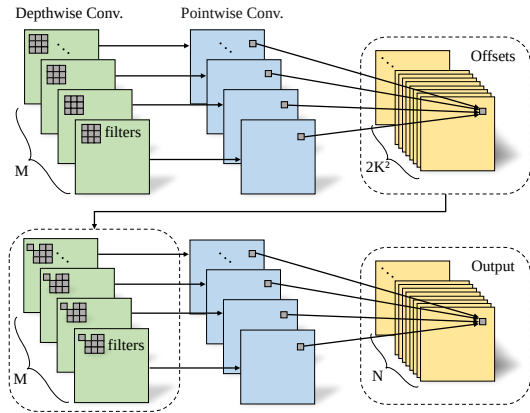


Figure 3: **An illustration of DSDCN.** The offsets are generated by $K \times K$ depthwise convolutions and pointwise convolutions, and the output is generated by $K \times K$ depthwise deformable convolutions and pointwise convolutions.

patch embedding to generate multi-scale tokens, and then feed them into multiple Transformer branches, respectively. Each Transformer branch contains multiple Transformer encoders. We apply the SKFF [79] module at the end of the multi-branch Transformer block to fuse features generated by different branches. Benefiting from the excellent performance of the Taylorformer and multi-branch design, we have the opportunity to compress the number of channels and Transformer blocks. We apply pixel-unshuffle and pixel-shuffle operations [53] at each stage to down-sample and up-sample features, respectively. Skip-connection [51] is used to aggregate information from encoder and decoder, and a 1×1 convolutional layer is employed for dimensionality reduction (except for the first stage). We also use a residual block after the encoder-decoder structure to restore fine structural and textural details. Finally, a 3×3 convolutional layer is applied to reduce channels and output a residual image $R \in \mathbb{R}^{3 \times h \times w}$. We thus obtain the restored image by $I' = I + R$. To further compress the computation and parameters, we apply depthwise separable convolutions [23, 11] in the model.

3.2. Multi-scale Patch Embedding

Visual elements can vary greatly in scale. Existing works [73, 35, 78] employ convolution with fixed kernels into patch embedding, which may result in single scale of visual tokens. To address this issue, we design a new multi-scale patch embedding with three properties: 1) multiple sizes of receptive field, 2) multi-level semantic information, 3) flexible shapes of receptive field. Specifically, by designing multiple deformable convolutions (DCN) [13] with different scales of convolution kernels in parallel, we enable patch embedding to generate both coarse and fine visual tokens, as well as flexible transformation modeling

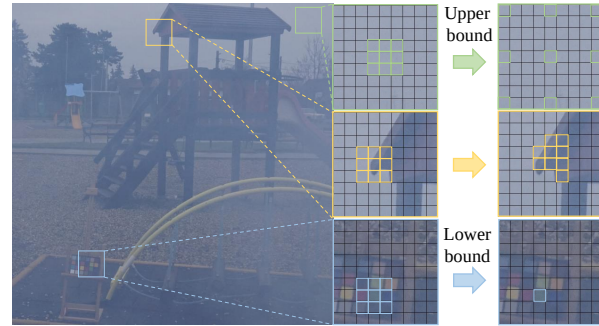


Figure 4: **An illustration of the receptive field of DSDCN (the offsets are truncated to [-3,3]).** The upper bound of the receptive field of the DSDCN is 9×9 and the lower bound is 1×1 .

capabilities. Inspired by the operation of stacking conventional layers that can expand receptive fields [56], we stack several deformable convolutional layers with small kernels instead of a single deformable convolutional layer with large kernels. This not only increases the depth of network and consequently provides multi-level semantic information, but also helps to reduce parameters and computational burden. All deformable convolutional layers are followed by Hardswish [22] activation functions.

Similar to the strategy of depthwise separable convolutions [23, 11], we propose depthwise separable and deformable convolutions (DSDCN), which decomposes the parts of DCN with depthwise convolution and pointwise convolution, as shown in Fig. 3. The computational cost of standard DCN and DSDCN regarding a $h \times w$ image is as follows:

$$\Omega(\text{DCN}) = 2MK^4hw + MNK^2hw + 4MK^2hw, \quad (1)$$

$$\Omega(\text{DSDCN}) = 8MK^2hw + MNhw. \quad (2)$$

where M and N are the numbers of channels in the input and output, respectively, and K is the kernel size of convolution. The number of parameters of DCN and DSDCN are as follows:

$$P(\text{DCN}) = 2MK^4 + MNK^2, \quad (3)$$

$$P(\text{DSDCN}) = 4MK^2 + MN, \quad (4)$$

In summary, DSDCN greatly reduces computational complexity and number of parameters compared to DCN.

Considering that images are locally relevant and patch embedding captures the basic elements of feature maps, the visual elements (*i.e.*, tokens) should be more focused on local areas. We control the receptive field range of the patch embedding layer by truncating the offsets, which we choose in practice to be $[-3, 3]$. As it is shown in Fig. 4, depending

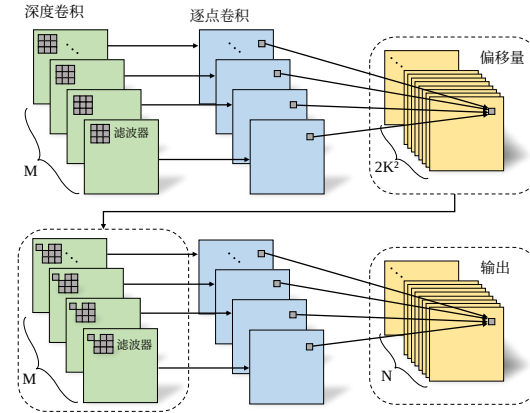


图3: **DSDCN 的示意图。**偏移量由 $K \times K$ 深度卷积和逐点卷积生成，输出由 $K \times K$ 深度可变形卷积和逐点卷积生成。

块嵌入生成多尺度令牌，然后分别将其馈入多个 Transformer 分支。每个 Transformer 分支包含多个 Transformer 编码器。我们在多分支 Transformer 块的末端应用 SKFF [79] 模块，以融合不同分支生成的特征。得益于 TaylorFormer 和多分支设计的优异性能，我们有机会压缩通道数和 Transformer 块的数量。我们在每个阶段分别应用像素反重排和像素重排操作 [53] 来下采样和上采样特征。跳跃连接 [51] 用于聚合来自编码器和解码器的信息，并使用一个 1×1 卷积层进行降维（第一阶段除外）。我们还在编码器-解码器结构之后使用残差块来恢复精细的结构和纹理细节。最后，应用一个 3×3 卷积层来减少通道数并输出残差图像 $R \in \mathbb{R}^{3 \times h \times w}$ 。由此，我们通过 $I' = I + R$ 得到恢复图像。为了进一步压缩计算量和参数量，我们在模型中应用了深度可分离卷积 [23, 11]。

3.2. 多尺度块嵌入

视觉元素的尺度变化可能非常大。现有工作 [73, 35, 78] 在块嵌入中使用固定卷积核的卷积，这可能导致视觉令牌只有单一尺度。为了解决这个问题，我们设计了一种新的多尺度块嵌入，具有三个特性：1) 多种感受野尺寸，2) 多级语义信息，3) 灵活的感受野形状。具体来说，通过并行设计多个具有不同尺度卷积核的可变形卷积 (DCN) [13]，我们使块嵌入能够同时生成粗粒度和细粒度的视觉令牌，并具备灵活的变换建模

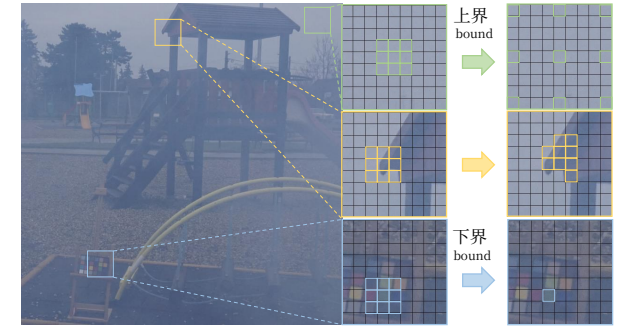


图4: **DSDCN 感受野的示意图（偏移量被截断至 [-3,3]）。**DSDCN 感受野的上界为 9×9 ，下界为 1×1 。

能力。受堆叠传统卷积层可以扩大感受野这一操作的启发 [56]，我们堆叠多个具有小卷积核的可变形卷积层，而不是使用单个具有大卷积核的可变形卷积层。这不仅增加了网络深度，从而提供多级语义信息，还有助于减少参数量和计算负担。所有可变形卷积层后都跟有 Hardswish [22] 激活函数。

类似于深度可分离卷积的策略 [23, 11]，我们提出了深度可分离可变形卷积 (DSDCN)，它将 DCN 的部分分解为深度卷积和逐点卷积，如图 3 所示。标准 DCN 和 DSDCN 关于一张 $h \times w$ 图像的计算成本如下：

$$\Omega(\text{DCN}) = 2MK^4hw + MNK^2hw + 4MK^2hw, \quad (1)$$

$$\Omega(\text{DSDCN}) = 8MK^2hw + MNhw. \quad (2)$$

其中 M 和 N 分别是输入和输出的通道数，K 是卷积的卷积核大小。DCN 和 DSDCN 的参数量如下：

$$P(\text{DCN}) = 2MK^4 + MNK^2, \quad (3)$$

$$P(\text{DSDCN}) = 4MK^2 + MN, \quad (4)$$

总之，与 DCN 相比，DSDCN 大幅降低了计算复杂度和参数量。

考虑到图像具有局部相关性，且块嵌入能够捕捉特征图的基本元素，视觉元素（即令牌）应更加关注局部区域。我们通过截断偏移量来控制块嵌入层的感受野范围，实践中我们选择其为 $[-3, 3]$ 。如图 4 所示，

on the shape of visual object, the model is able to select the receptive field size by learning, which has an upper bound of 9×9 , equivalent to a dilated convolution [77] of BF = 4, and a lower bound of 1×1 . When we set up multi-scale patch embedding in parallel, the sizes of the receptive field of different branches are $x \in [1, 9]$, $y \in [x, x + 8]$ and $z \in [y, y + 8]$ in ascending order (for three branches). Experiments in the supplementary material show that setting constraints on the receptive field of each token in a reasonable way can improve the model’s performance.

3.3. Taylor Expanded Multi-head Self-Attention

Let queries (Q), keys (K), and values (V) be a sequence of $h \times w$ feature vectors with dimensions D , where h and w is the height and width of the image, respectively. The formula of the vanilla Transformer [64] is as follows:

$$V' = \text{Softmax} \left(\frac{Q^T K}{\sqrt{D}} \right) V^T. \quad (5)$$

Since $Q \in \mathbb{R}^{hw \times D}$, $K \in \mathbb{R}^{hw \times D}$ and $V \in \mathbb{R}^{hw \times D}$, Softmax causes the computational complexity of self-attention to be $\mathcal{O}(h^2 w^2)$, resulting in expensive computational cost.

We aim to reduce the computational complexity of self-attention from $\mathcal{O}(h^2 w^2)$ to $\mathcal{O}(hw)$. To achieve it, we first write the generalized attention equation for Eq. (5), as follows:

$$V'_i = \frac{\sum_{j=1}^N f(Q_i, K_j) V_j}{\sum_{j=1}^N f(Q_i, K_j)}, \quad (6)$$

where the matrix with i as subscript is the vector of the i -th row of matrix, and $f(\cdot)$ denotes any similarity function. Eq. (6) degenerates to Eq. (5) when we let $f(Q_i, K_j) = \exp \left(\frac{Q_i^T K_j}{\sqrt{D}} \right)$. If we apply the Taylor formula to perform a first-order Taylor expansion on $\exp \left(\frac{Q_i^T K_j}{\sqrt{D}} \right)$ at 0, we can rewrite Eq. (6) as:

$$V'_i = \frac{\sum_{j=1}^N (1 + Q_i^T K_j + o(Q_i^T K_j)) V_j^T}{\sum_{j=1}^N (1 + Q_i^T K_j + o(Q_i^T K_j))}. \quad (7)$$

Further, we generate $\tilde{Q}_i, \tilde{K}_i$ from the normalization of vectors Q_i and K_j to approximate $\exp \left(\frac{Q_i^T K_j}{\sqrt{D}} \right)$. When the norm of $\tilde{Q}_i$ and $\tilde{K}_i$ are smaller than 1, we can make the values of the attention map all positive, and in practice, we find the best results are achieved by normalizing the norm to 0.5. As shown in Fig. 5, we consider that there is an approximation to e^x and its first-order Taylor expansion is in the definition domain of $[-0.25, 0.25]$. So we eliminate Peano’s form of remainder [43] and obtain the expression for the Taylor expansion of self-attention as follows:

$$V'_i = \frac{\sum_{j=1}^N (1 + \tilde{Q}_i^T \tilde{K}_j) V_j^T}{\sum_{j=1}^N (1 + \tilde{Q}_i^T \tilde{K}_j)}. \quad (8)$$

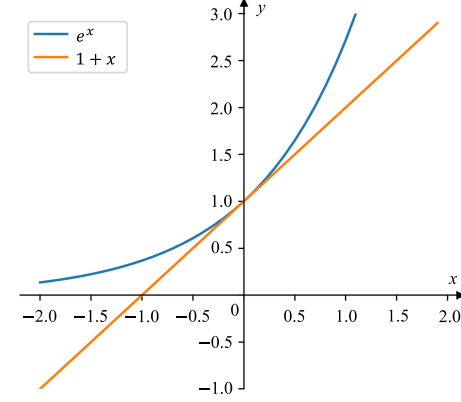


Figure 5: e^x (blue) and its first-order Taylor expansion curve (orange). The closer to 0, the tighter the approximation of the orange line to the blue line.

Finally, we apply the matrix multiplication associative law to Eq. (8), as follows:

$$\begin{aligned} V'_i &= \text{Taylor-Attention}(Q_i, K_i, V_i) \\ &= \frac{\sum_{j=1}^N V_j^T + \tilde{Q}_i^T \sum_{j=1}^N \tilde{K}_j V_j^T}{N + \tilde{Q}_i^T \sum_{j=1}^N \tilde{K}_j}. \end{aligned} \quad (9)$$

Supplementary materials provide the pseudo-code of Taylor expanded multi-head self-attention (T-MSA).

Similar to MDTA [78], we employ deep convolutional generation for Q, K, V to emphasize the local context. We also employ a multi-head structure, where the number of heads increases progressively from top to bottom of the level. The computational complexity of the standard multi-head self-attention (MSA) module and the T-MSA module regarding an image of $h \times w$ patches is as follows:

$$\Omega(\text{MSA}) = 4hwD^2 + 2h^2w^2D, \quad (10)$$

$$\Omega(\text{T-MSA}) = 18hwD + 7hwD^2. \quad (11)$$

Generally, $h \times w$ is much larger than D , so T-MSA provides more possibilities than MSA for testing high-resolution images, and ensures the values are close to the MSA.

3.4. Multi-scale Attention Refinement

Since we perform a first-order Taylor expansion of softmax in T-MSA and ignore Peano’s form of reminder, there is an inevitable approximation error. For the n -th order remainder term $\frac{(Q_i K_j)^n}{n!}$ ($n \geq 2$) of Taylor expansion, the matrix multiplicative combination law cannot be used to make the computational complexity of T-MSA linear. However, the remainder term is related to the Q and K matrices. Considering that the images have local correlation, we learn the local information of the Q and K matrices to correct the inaccurate output V' . In addition, the conv-attention module

根据视觉对象的形状，模型能够通过学习来选择感受野大小，其上界为 9×9 ，相当于膨胀系数为 [77] 的BF = 4的空洞卷积，下界为 1×1 。当我们并行设置多尺度块嵌入时，不同分支的感受野大小依次为 $x \in [1, 9]$, $y \in [x, x + 8]$ 和 $z \in [y, y + 8]$ （对于三个分支，按升序排列）。补充材料中的实验表明，以合理方式对每个令牌的感受野设置约束可以提升模型的性能。

3.3. 泰勒展开多头自注意力

设查询 (Q)、键 (K) 和值 (V) 为维度为 D 的 $h \times w$ 个特征向量序列，其中 h 和 w 分别为图像的高度和宽度。原始Transformer [64] 的公式如下：

$$V' = \text{Softmax} \left(\frac{Q^T K}{\sqrt{D}} \right) V^T. \quad (5)$$

由于 $Q \in \mathbb{R}^{hw \times D}$ 、 $K \in \mathbb{R}^{hw \times D}$ 和 $V \in \mathbb{R}^{hw \times D}$ ，Softmax 导致自注意力的计算复杂度为 $\mathcal{O}(h^2 w^2)$ ，从而带来高昂的计算成本。

我们旨在将自注意力的计算复杂度从 $\mathcal{O}(h^2 w^2)$ 降低到 $\mathcal{O}(hw)$ 。为此，我们首先将式 (5) 的广义注意力方程写为如下形式：

$$V'_i = \frac{\sum_{j=1}^N f(Q_i, K_j) V_j}{\sum_{j=1}^N f(Q_i, K_j)}, \quad (6)$$

其中以 i 为下标的矩阵表示矩阵的第 i 行向量，而 $f(\cdot)$ 表示任意相似度函数。当令 $f(Q_i, K_j) = \exp \left(\frac{Q_i^T K_j}{\sqrt{D}} \right)$ 时，式 (6) 退化为式 (5)。如果我们应用泰勒公式对 $\left(\frac{Q_i^T K_j}{\sqrt{D}} \right)$ 处的 $\exp$ 进行一阶泰勒展开，则可将式 (6) 改写为：

$$V'_i = \frac{\sum_{j=1}^N (1 + Q_i^T K_j + o(Q_i^T K_j)) V_j^T}{\sum_{j=1}^N (1 + Q_i^T K_j + o(Q_i^T K_j))}. \quad (7)$$

进一步地，我们生成 $\tilde{Q}_i, \tilde{K}_i$ 来自向量 Q_i 和 K_j 的归一化，以近似 $\exp \left(\frac{Q_i^T K_j}{\sqrt{D}} \right)$ 。当 $\tilde{Q}_i, \tilde{K}_i$ 的范数 Q_i 和 K_i 小于1时，我们可以使注意力图的值全部为正，并且在实践中，我们发现将范数归一化到0.5可以获得最佳结果。如图5所示，我们认为在 e^x 的定义域内存在对其的近似，且它的一阶泰勒展开位于 $[-0.25, 0.25]$ 的定义域内。因此，我们消除佩亚诺余项 [43]，并得到自注意力泰勒展开的表达式如下：

$$V'_i = \frac{\sum_{j=1}^N (1 + \tilde{Q}_i^T \tilde{K}_j) V_j^T}{\sum_{j=1}^N (1 + \tilde{Q}_i^T \tilde{K}_j)}. \quad (8)$$

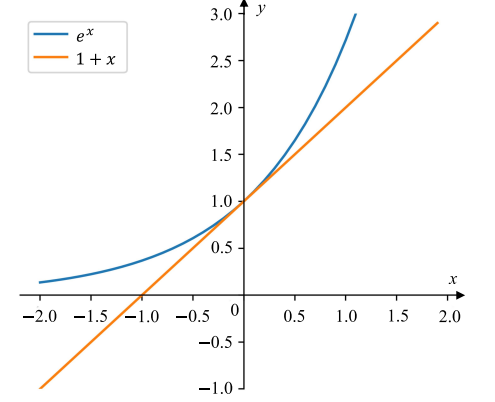


图5: e^x (蓝色) 及其一阶泰勒展开曲线 (橙色)。越接近0，橙色线对蓝色线的逼近越紧密。

最后，我们将矩阵乘法结合律应用于式 (8)，如下所示：

$$\begin{aligned} V'_i &= \text{Taylor-Attention}(Q_i, K_i, V_i) \\ &= \frac{\sum_{j=1}^N V_j^T + \tilde{Q}_i^T \sum_{j=1}^N \tilde{K}_j V_j^T}{N + \tilde{Q}_i^T \sum_{j=1}^N \tilde{K}_j}. \end{aligned} \quad (9)$$

补充材料提供了泰勒展开多头自注意力 (T-MSA) 的伪代码。

与MDTA类似 [78]，我们采用深度卷积生成 Q, K, V 来强调局部上下文。我们还采用了多头结构，其中头数从层级顶部到底部逐级增加。对于具有 $h \times w$ 个图像块的图像，标准多头自注意力 (MSA) 模块和 T-MSA模块的计算复杂度如下：

$$\Omega(\text{MSA}) = 4hwD^2 + 2h^2w^2D, \quad (10)$$

$$\Omega(\text{T-MSA}) = 18hwD + 7hwD^2. \quad (11)$$

通常， $h \times w$ 远大于 D ，因此T-MSA在测试高分辨率图像时比MSA提供了更多可能性，并确保其值与MSA接近。

3.4. 多尺度注意力细化

由于我们在T-MSA中对softmax进行了一阶泰勒展开，并忽略了皮亚诺余项，因此不可避免地存在近似误差。对于泰勒展开的 n 阶余项 $\frac{(Q_i K_j)^n}{n!}$ ($n \geq 2$)，矩阵乘法结合律无法用于使T-MSA的计算复杂度线性化。然而，余项与Q和K矩阵相关。考虑到图像具有局部相关性，我们学习Q和K矩阵的局部信息来校正不准确的输出 V' 。此外，卷积注意力模块

Table 1: **Quantitative comparisons of various methods on dehazing benchmarks.**“-” indicates that the result is not available. The best and second best results are highlighted in bold and underlined, respectively.

Methods	SOTS-Indoor		SOTS-Outdoor		O-HAZE		Dense-Haze		Overhead	
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	#Param	MACs
DCP [20]	16.62	0.818	19.13	0.815	16.78	0.653	12.72	0.442	-	0.6G
DehazeNet [5]	19.82	0.821	24.75	0.927	17.57	0.770	13.84	0.430	0.01M	0.6G
GFN [50]	22.30	0.880	21.55	0.844	18.16	0.671	-	-	0.50M	14.9G
GDN [37]	32.16	0.984	30.86	0.982	18.92	0.672	14.96	0.536	0.96M	21.5G
PFDN [15]	32.68	0.976	-	-	-	-	-	-	11.27M	51.5G
MSBDN [14]	33.67	0.985	33.48	0.982	24.36	0.749	15.13	0.555	31.35M	41.54G
FFA-Net [45]	36.39	0.989	33.57	0.984	22.12	0.770	15.70	0.549	4.46M	287.8G
AECR-Net [74]	37.17	0.990	-	-	-	-	15.80	0.466	2.61M	52.2G
MAXIM-2S [62]	38.11	0.991	34.19	0.985	-	-	-	-	14.10M	216.0G
SGID-PFF [4]	38.52	0.991	30.20	0.975	20.96	0.741	12.49	0.517	13.87M	156.4G
Restormer [78]	38.88	0.991	-	-	23.58	0.768	15.78	0.548	26.10M	141.0G
Dehamer [19]	36.63	0.988	35.18	0.986	<u>25.11</u>	0.777	<u>16.62</u>	<u>0.560</u>	132.50M	60.3G
Ours (-B)	<u>40.71</u>	<u>0.992</u>	<u>37.42</u>	<u>0.989</u>	25.05	0.788	16.66	<u>0.560</u>	2.68M	38.5G
Ours (-L)	42.64	0.994	38.09	0.991	25.31	<u>0.782</u>	16.44	0.566	7.43M	88.1G

allows TaylorFormer to better handle high frequency information [42].

Specifically, for multi-heads $Q_m \in \mathbb{R}^{head \times \frac{D}{head} \times N}$ and $K_m \in \mathbb{R}^{head \times \frac{D}{head} \times N}$, we reshape them into $\hat{Q}_m \in \mathbb{R}^{head \times \frac{D}{head} \times H \times W}$ and $\hat{K}_m \in \mathbb{R}^{head \times \frac{D}{head} \times H \times W}$, where $head$ denotes the number of heads, and we concatenate $\hat{Q}_m$ and $\hat{K}_m$ along the channel dimension to generate a tensor $T \in \mathbb{R}^{head \times \frac{2D}{head} \times H \times W}$, which is subsequently passed through a multi-scale grouped convolutional layer to generate a gating tensor $G \in \mathbb{R}^{head \times 1 \times H \times W}$ as:

$$G = \text{Sigmoid} \left(\text{Concat} \left(T_1 W_1^G, \dots, T_{head} W_{head}^G \right) \right), \quad (12)$$

where $T_{head} \in \mathbb{R}^{\frac{D}{head} \times H \times W}$ is the $head$ -th head of T , and $W_{(\cdot)}^G$ is the convolution with different kernels. Since different levels of the network have different numbers of heads, we choose the corresponding multi-scale grouped convolutions for different numbers of heads. Supplementary materials provide details of the structure of the MSAR module.

With the approach of T-MSA and the module of MSAR, the refined T-MSA module is computed as:

$$\hat{X} = X + \text{Cat} \left(H_1 \odot G_1, \dots, H_{head} \odot G_{head} \right) W^P, \\ H_i = \text{Taylor-Attention} \left(QW_i^Q, KW_i^K, VW_i^V \right), \quad (13)$$

where X and $\hat{X}$ denote the input and output feature maps. The projections are parameter matrices $W^P \in \mathbb{R}^{D \times D}$, $W_i^Q \in \mathbb{R}^{D \times \frac{D}{head}}$, $W_i^K \in \mathbb{R}^{D \times \frac{D}{head}}$, and $W_i^V \in \mathbb{R}^{D \times \frac{D}{head}}$.

4. Experiments

In this section, we conduct experiments to demonstrate the effectiveness of the proposed MB-TaylorFormer. Further details including more qualitative results are provided in Supplementary Materials.

4.1. Experiment Setup

Implementation Details. We provide two architectures of MB-TaylorFormer including MB-TaylorFormer-B (the basic model) and MB-TaylorFormer-L (a larger variant). We employ random cropping and random flipping for data augmentation. We set the initial learning rate to 2e-4 and gradually reduce it to 1e-6 using cosine annealing [40]. we only use L1 loss as our loss function.

Datasets. We evaluate the proposed MB-TaylorFormer on synthetic dataset (RESIDE [32]) and real-world datasets (O-HAZE [2], Dense-Haze [1]). As subsets of RESIDE, ITS and OTS contain 13990 pairs of indoor and 313950 pairs of outdoor images, respectively. Models are evaluated on the SOTS subset. The real-world datasets, O-HAZE and Dense-Haze, contain 45 and 55 paired images, respectively. We use the last 5 images of each dataset as the testing set and the rest as the training set.

4.2. Experiments on Synthetic Hazy Images

Table 1 compares the performance of MB-TaylorFormer with the SOTA methods on synthetic datasets. Our basic model (MB-TaylorFormer-B) achieves 40.71dB PSNR and 0.994 SSIM on SOTS-Indoor. It improves the PSNR by 2.19dB over the previous SOTA method SGID-PFF [4], but with merely 10% number of parameters and 14% computational cost of SGID-PFF. Furthermore, our large model (MB-TaylorFormer-L) achieves 4.12dB gain over the SGID-PFF with approximately half of its complexity. Our method also surpasses the second best performing method Dehamer [19] by 2.91dB on the SOTS-Outdoor. We also compare the visual results of MB-TaylorFormer-L with other SOTA dehazing methods. Fig. 6 shows that the images generated by other methods are less natural in the shadows and high frequency regions. However, our

表1: 各种方法在去雾基准数据集上的定量比较。“-”表示结果不可用。最佳和次佳结果分别以粗体和下划线突出显示。

方法	SOTS-Indoor		SOTS-Outdoor		O-HAZE		浓雾		开销	
	峰值信噪比↑	结构相似性↑	峰值信噪比↑	结构相似性↑	峰值信噪比↑	结构相似性↑	峰值信噪比↑	结构相似性↑	参数量	MACs
暗通道先验 [20]	16.62	0.818	19.13	0.815	16.78	0.653	12.72	0.442	-	0.6G
DehazeNet [5]	19.82	0.821	24.75	0.927	17.57	0.770	13.84	0.430	0.01M	0.6G
GFN [50]	22.30	0.880	21.55	0.844	18.16	0.671	-	-	0.50M	14.9G
GDN [37]	32.16	0.984	30.86	0.982	18.92	0.672	14.96	0.536	0.96M	21.5G
PFDN [15]	32.68	0.976	-	-	-	-	-	-	11.27M	51.5G
MSBDN [14]	33.67	0.985	33.48	0.982	24.36	0.749	15.13	0.555	31.35M	41.54G
FFA-Net [45]	36.39	0.989	33.57	0.984	22.12	0.770	15.70	0.549	4.46M	287.8G
AECR-Net [74]	37.17	0.990	-	-	-	-	15.80	0.466	2.61M	52.2G
MAXIM-2S [62]	38.11	0.991	34.19	0.985	-	-	-	-	14.10M	216.0G
SGID-PFF [4]	38.52	0.991	30.20	0.975	20.96	0.741	12.49	0.517	13.87M	156.4G
Restormer [78]	38.88	0.991	-	-	23.58	0.768	15.78	0.548	26.10M	141.0G
Dehamer [19]	36.63	0.988	35.18	0.986	<u>25.11</u>	0.777	<u>16.62</u>	<u>0.560</u>	132.50M	60.3G
本文方法 (-B)	<u>40.71</u>	<u>0.992</u>	<u>37.42</u>	<u>0.989</u>	25.05	0.788	16.66	<u>0.560</u>	2.68M	38.5G
本文方法 (-L)	42.64	0.994	38.09	0.991	25.31	<u>0.782</u>	16.44	0.566	7.43M	88.1G

使TaylorFormer能够更好地处理高频信息 [42]。

具体来说，对于多头 $Q_m \in \mathbb{R}^{head \times \frac{D}{head} \times N}$ 和 $K_m \in \mathbb{R}^{head \times \frac{D}{head} \times N}$ ，我们将它们重塑为 $\hat{Q}_m \in \mathbb{R}^{head \times \frac{D}{head} \times H \times W}$ 和 $\hat{K}_m \in \mathbb{R}^{head \times \frac{D}{head} \times H \times W}$ ，

其中 $head$ 表示头的数量，并沿通道维度拼接 $\hat{Q}_m$ 和 $\hat{K}_m$ 以生成张量 $T \in \mathbb{R}^{head \times \frac{2D}{head} \times H \times W}$ ，随后通过多尺度分组卷积层生成门控张量 $G \in \mathbb{R}^{head \times 1 \times H \times W}$ ，如下所示：

$$G = \text{Sigmoid} \left(\text{Concat} \left(T_1 W_1^G, \dots, T_{head} W_{head}^G \right) \right), \quad (12)$$

其中 $T_{head} \in \mathbb{R}^{\frac{D}{head} \times H \times W}$ 是 T 的第 $head$ 个头， $W_{(\cdot)}^G$ 是具有不同卷积核的卷积。由于网络的不同层级具有不同的头数，我们针对不同的头数选择相应的多尺度分组卷积。补充材料提供了MSAR模块结构的详细信息。

借助 T-MSA 方法和 MSAR 模块，精炼后的 T-MSA 模块计算如下：

$$\hat{X} = X + \text{Cat} \left(H_1 \odot G_1, \dots, H_{head} \odot G_{head} \right) W^P, \\ H_i = \text{Taylor-Attention} \left(QW_i^Q, KW_i^K, VW_i^V \right), \quad (13)$$

其中 X 和 $\hat{X}$ 分别表示输入和输出特征图。投影为参数矩阵 $W^P \in \mathbb{R}^{D \times D}$ 、 $W_i^Q \in \mathbb{R}^{D \times \frac{D}{head}}$ 、 $W_i^K \in \mathbb{R}^{D \times \frac{D}{head}}$ 和 $W_i^V \in \mathbb{R}^{D \times \frac{D}{head}}$ 。

4. 实验

在本节中，我们通过实验来验证所提出的 MB-TaylorFormer 的有效性。更多细节（包括更多定性结果）见补充材料。

4.1. 实验设置

实现细节。我们提供了 MB-TaylorFormer 的两种架构，包括 MB-TaylorFormer-B（基础模型）和 MB-TaylorFormer-L（更大的变体）。我们采用随机裁剪和随机翻转进行数据增强。我们将初始学习率设为 2e-4，并使用余弦退火 [40] 将其逐渐降至 1e-6。我们仅使用 L1 损失作为损失函数。

数据集。我们在合成数据集（RESIDE [32]）和真实世界数据集（O-HAZE [2]、浓雾 [1]）上评估所提出的 MB-TaylorFormer。作为 RESIDE 的子集，ITS 和 OTS 分别包含 13990 对室内图像和 313950 对室外图像。模型在 SOTS 子集上进行评估。真实世界数据集 O-HAZE 和浓雾分别包含 45 对和 55 对配对图像。我们使用每个数据集的最后 5 张图像作为测试集，其余图像作为训练集。

4.2. 合成雾天图像上的实验

表 1 比较了 MB-TaylorFormer 与最先进方法在合成数据集上的性能。我们的基础模型（MB-TaylorFormer-B）在 SOTS-Indoor 上达到了 40.71dB 峰值信噪比和 0.994 结构相似性。与之前的最先进方法 SGID-PFF 相比，其峰值信噪比提高了 2.19dB [4]，但参数量仅为 SGID-PFF 的 10%，计算成本仅为 SGID-PFF 的 14%。此外，我们的较大模型（MB-TaylorFormer-L）以约一半的复杂度实现了比 SGID-PFF 高 4.12dB 的增益。我们的方法在 SOTS-Outdoor 上还比排名第二的方法 Dehamer [19] 高出 2.91dB。我们还将 MB-TaylorFormer-L 与其他最先进去雾方法的视觉结果进行了比较。图 6 显示，其他方法生成的图像在阴影和低频区域不够自然。然而，我们的

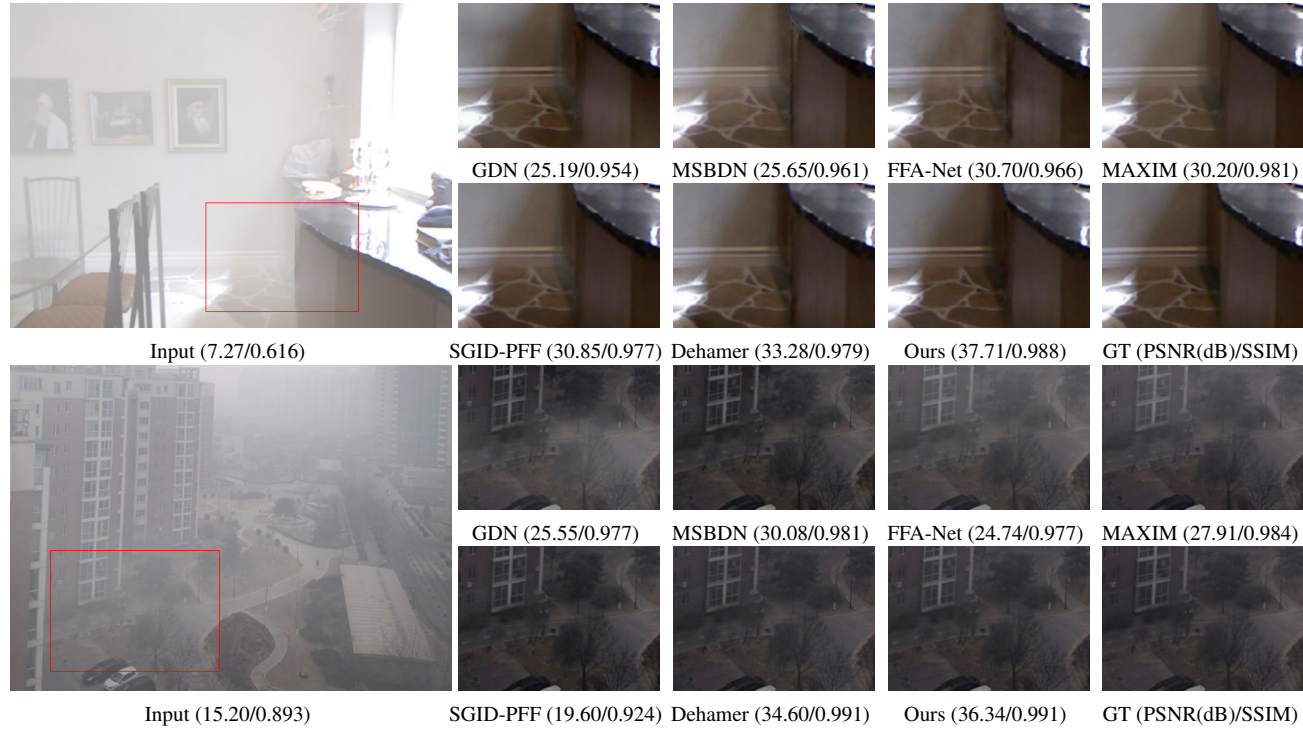


Figure 6: **Visual comparisons on synthetic hazy images.** The first two rows of images are from the SOTS-Indoor dataset, the last two rows are from SOTS-Outdoor. Our MB-TaylorFormer-L generates haze-free images with color fidelity and finer textures.

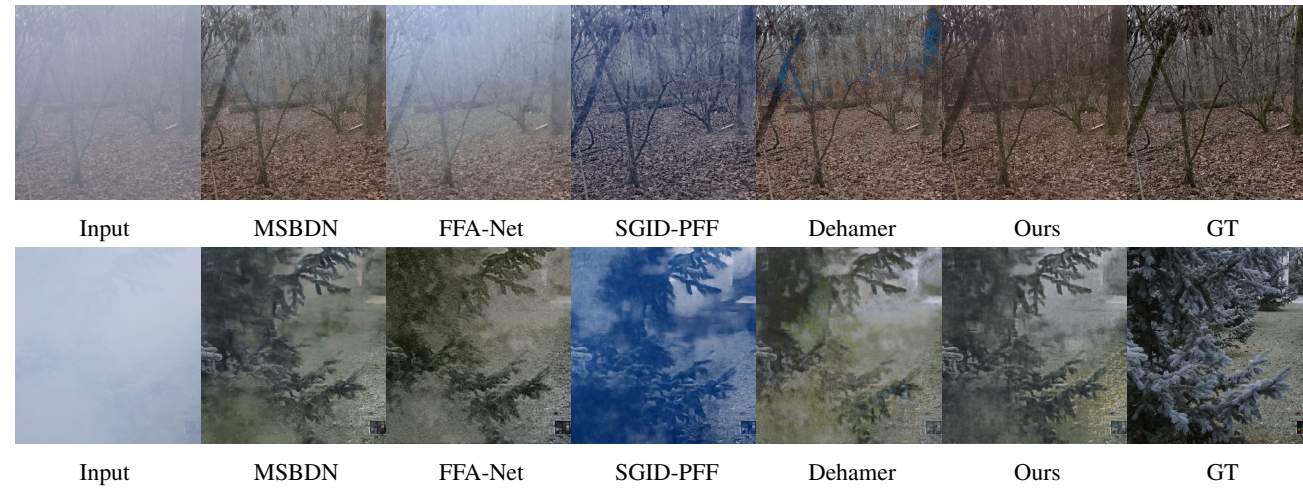


Figure 7: **Visual comparisons on real hazy images.** The first row of images is from the O-HAZE, The second row of images is from the Dense-Haze. Our MB-TaylorFormer generates haze-free images with color fidelity and without artifacts.

MB-TaylorFormer generates haze-free images with better-restored details in the shadows, and is most similar to ground truth, especially in the high-frequency region.

4.3. Experiments on Real Hazy Images

We further evaluate our MB-TaylorFormer against the SOTA methods on the real datasets O-HAZE [2] and Dense-Haze [1]. The quantitative comparison is shown in Table 1.

We have the following findings: 1) MB-TaylorFormer outperforms previous methods by up to 0.20dB and 0.04dB PSNR magnitude on the O-HAZE and Dense-Haze, respectively. 2) Usually, when training on small-size datasets, the non-convex losses of visual Transformer are expected to lead to poor performance [42]. However, MB-TaylorFormer still achieves the best PSNR and SSIM scores than other CNN-based models on small-size datasets. We also com-



图 6: 合成雾天图像的视觉对比。前两行图像来自 SOTS-Indoor 数据集, 后两行来自 SOTS-Outdoor。我们的 MB-TaylorFormer-L 生成了具有色彩保真度和更精细纹理的无雾图像。

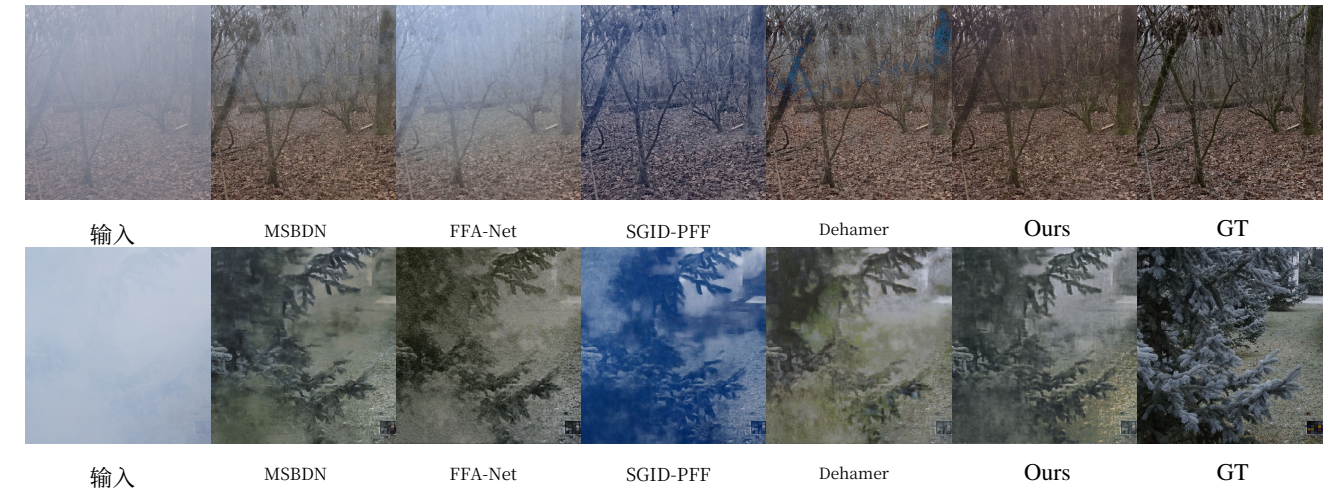


图7: 真实雾图的视觉效果对比。第一行图像来自O-HAZE, 第二行图像来自Dense-Haze。我们的 MB-TaylorFormer能够生成色彩保真且无伪影的无雾图像。

MB-TaylorFormer生成的无雾图像在阴影中恢复了更好的细节, 并且与真值最为相似, 尤其是在高频区域。

4.3. 真实雾图实验

我们进一步在真实数据集O-HAZE [2]和Dense-Haze [1]上我们的MB-TaylorFormer与最先进方法进行了评估。定量比较如表1所示。

我们有以下发现: 1) MB-TaylorFormer在O-HAZE和Dense-Haze数据集上分别以高达0.20dB和0.04dB的PSNR幅度超越了先前的方法。2) 通常, 在小规模数据集上训练时, 视觉Transformer的非凸损失预期会导致性能不佳[42]。然而, MB-TaylorFormer在小规模数据集上仍然取得了优于其他基于CNN的模型的最佳PSNR和SSIM分数。我们还比-

Table 2: **Ablation studies for the multi-scale patch embedding and multi-branch structure.** “-S” means two convolutional layers in series with the same kernel size of 3, and “-P” means two convolutional layers in parallel with the same kernel size of 3. Supplementary materials provide more details.

Branch	Type of Conv.	PSNR	SSIM	#Params	MACs
Single	Conv	38.27	0.991	2.655M	33.63G
Double	Conv-P	38.42	0.991	2.652M	37.89G
	Dilated Conv-P	38.77	0.991	2.652M	37.89G
	Conv-S	39.04	0.992	2.652M	37.89G
	DSDCN-S	40.71	0.992	2.677M	38.51G

Table 3: **Effectiveness of multi-scale attention refinement module.** It balances computational burden and performance.

Methods	PSNR	SSIM	#Params	MACs
MB-TaylorFormer-B	40.71	0.992	2.677M	38.51G
W/o MSAR	38.74	0.990	2.582M	37.60G
$G: \mathbb{R}^{h \times 1 \times H \times W} \rightarrow \mathbb{R}^{h \times C \times H \times W}$	40.88	0.993	4.288M	68.43G

pare our MB-TaylorFormer with previous methods in terms of the visual quality of dehazed images, which are presented in Fig. 7. The dehazed images by MSBDN [14] hardly remove the haze, FFA-Net [45] generates images with graininess and loss of detail, SGID-PFF [4] generates images with color distortion, and the result images by Dehamer [19] suffer from artifacts and texture loss. The haze-free images generated by our method are much cleaner.

4.4. Ablation Studies

In this section, all MB-TaylorFormer models are trained on 256×256 pixel patches cropped from SOTS-Indoor. The epoch is set as 500. Based on MB-TaylorFormer-B, we analyze the effectiveness of different modules on our framework. MB-TaylorFormer-L is used to explore the effects of branch dimension, channel dimension, and depth.

Study of multi-scale patch embedding and multi-branch structure. In Table 2, we study the difference in the patch embedding and the different number of branches. Specifically, we set a single-branch model based on single-scale standard convolution as the baseline and modify it from the following aspects. 1) To examine the effect of multi-branch structure, we design the patch embedding on models with single-scale and multi-branching (Conv-P). 2) To study the effect of multiple receptive field sizes, we use parallel dilated convolutional layers (DF=1, 2) to embed patches (Dilated Conv-P). 3) To investigate the influence of multi-level semantic information, we further replace dilated convolution with standard convolution to embed patches, and employ the approach of connecting two convolutional layers in series (Conv-S). 4) To examine the effect of flex-

Table 4: **Comparison with other linear self-attention modules.** We replace the T-MSA of our proposed model with another linear self-attention module.

Methods	PSNR	SSIM	#Params	MACs
MB-TaylorFormer-B	40.71	0.992	2.677M	38.51G
W/o MSAR	38.74	0.990	2.582M	37.60G
T-MSA $\rightarrow$ MDTA [78]	38.57	0.991	2.582M	35.20G
T-MSA $\rightarrow$ Swin [35]	36.59	0.988	2.517M	36.38G
T-MSA $\rightarrow$ TAR [28]	36.74	0.987	2.517M	34.00G
T-MSA $\rightarrow$ PVTv2 [72]	38.10	0.990	10.89M	38.60G
T-MSA $\rightarrow$ Cswin [16]	38.19	0.987	3.30M	40.10G
T-MSA $\rightarrow$ LinFormer [68]	36.12	0.983	48.70M	371.68G

Table 5: **Analysis of approximation errors.** The smaller approximation error for softmax-attention, the better the performance.

Methods	PSNR	SSIM	#Params	MACs
Swin	36.59	0.988	2.517M	36.38G
Swin + T-MSA-2nd	36.50	0.988	2.517M	36.38G
Swin + T-MSA-1st	36.37	0.987	2.517M	36.38G

ible receptive field shapes, we additionally replace standard convolution with DSDCN (DSDCN-S), as shown in Fig. 2(b). The experiment shows that the performance from best to worst is DSDCN-S, Conv-S, Dilated Conv-S, Conv-P and Conv. This indicates that our multi-scale patch embedding can embed patches flexibly. As shown in Fig. 8, we visualize the average of the feature maps in the first stage of our network, and find that the multi-scale tokens (DSDCN-S) help the network to obtain richer information than the single-scale tokens (Conv). Thanks to the depth-separable design, our DSDCN has only a tiny increase in the number of parameters and computational cost compared to the depth-separable convolution.

Effectiveness of multi-scale attention refinement module. Table 3 demonstrates our MSAR module provides a favorable gain of 1.97dB over the counterpart without MSAR module, with only a tiny increase in the number of parameters (0.1M) and MACs (0.905G). When we set the gating tensor $G \in \mathbb{R}^{h \times 1 \times H \times W} \rightarrow G \in \mathbb{R}^{h \times C \times H \times W}$, it provides only a tiny improvement in PSNR and SSIM, at the cost of a significant increase in the number of parameters and computation complexity. Taking into account both performance and computational burden, we set $G \in \mathbb{R}^{h \times 1 \times H \times W}$.

Comparison with other linear self-attention modules. Table 4 compares the proposed T-MSA with several common linear self-attentive modules. Results show that the TaylorFormer has significant advantages over existing linear self-attention modules, which is attributed to the ability of T-MSA to model long-distance pixels and the approximation to the softmax-attention.

Analysis of approximation errors. To explore the approximation error and its impact, we investigate the ef-

表2: 针对多尺度分块嵌入与多分支结构的消融实验。

“-S”表示两个卷积层串联, 卷积核大小均为3; “-P”表示两个卷积层并联, 卷积核大小均为3。补充材料提供了更多细节。

分支	卷积类型	PSNR	SSIM	参数量	MACs
Single	Conv	38.27	0.991	2.655M	33.63G
双	Conv-P	38.42	0.991	2.652M	37.89G
	扩张卷积-P	38.77	0.991	2.652M	37.89G
	Conv-S	39.04	0.992	2.652M	37.89G
	DSDCN-S	40.71	0.992	2.677M	38.51G

表3: 多尺度注意力细化模块的有效性。它在计算负担与性能之间取得了平衡。

方法	PSNR	SSIM	参数量	MACs
MB-TaylorFormer-B	40.71	0.992	2.677M	38.51G
无MSAR	38.74	0.990	2.582M	37.60G
$G: \mathbb{R}^{h \times 1 \times H \times W} \rightarrow \mathbb{R}^{h \times C \times H \times W}$	40.88	0.993	4.288M	68.43G

较了我们的MB-TaylorFormer与先前方法在去雾图像视觉质量方面的表现, 结果如图7所示。MSBDN生成的无雾图像 [14] 几乎无法去除雾气, FFA-Net [45] 生成的图像存在颗粒感和细节丢失, SGID-PFF [4] 生成的图像存在颜色失真, 而Dehamer生成的图像 [19]存在伪影和纹理损失。我们的方法生成的无雾图像则干净得多。

4.4. 消融实验

在本节中, 所有 MB-TaylorFormer 模型均在从 SOTS-Indoor 裁剪的 256×256 像素图像块上进行训练。训练轮数设置为 500。基于 MB-TaylorFormer-B, 我们分析了不同模块在我们的框架上的有效性。MB-TaylorFormer-L 用于探索分支维度、通道维度和深度的影响。

多尺度块嵌入与多分支结构研究。在表2中, 我们研究了块嵌入的差异以及不同分支数量的影响。具体来说, 我们将基于单尺度标准卷积的单分支模型设为基线模型, 并从以下几个方面对其进行修改。1) 为了考察多分支结构的效果, 我们在单尺度和多分支 (Conv-P) 模型上设计块嵌入。2) 为了研究多感受野尺寸的影响, 我们使用并行空洞卷积层 (DF=1, 2) 来嵌入图像块 (扩张卷积-P)。3) 为了探究多层级语义信息的影响, 我们进一步用标准卷积替代空洞卷积来嵌入图像块, 并采用串联两个卷积层的方式 (Conv-S)。4) 为了考察灵活的

表 4: 与其他线性自注意力模块的比较。我们用另一个线性自注意力模块替换了我们提出的模型中的 T-MSA。

方法	峰值信噪比	结构相似性	参数量	MACs
MB-TaylorFormer-B	40.71	0.992	2.677M	38.51G
无MSAR	38.74	0.990	2.582M	37.60G
T-MSA $\rightarrow$ MDTA [78]	38.57	0.991	2.582M	35.20G
T-MSA $\rightarrow$ Swin [35]	36.59	0.988	2.517M	36.38G
T-MSA $\rightarrow$ TAR [28]	36.74	0.987	2.517M	34.00G
T-MSA $\rightarrow$ PVTv2 [72]	38.10	0.990	10.89M	38.60G
T-MSA $\rightarrow$ Cswin [16]	38.19	0.987	3.30M	40.10G
T-MSA $\rightarrow$ LinFormer [68]	36.12	0.983	48.70M	371.68G

表5: 近似误差分析。softmax注意力的近似误差越小, 性能越好。

方法	PSNR	SSIM	参数量	MACs
Swin	36.59	0.988	2.517M	36.38G
Swin + T-MSA-2nd	36.50	0.988	2.517M	36.38G
Swin + T-MSA-1st	36.37	0.987	2.517M	36.38G

感受野形状的效果, 我们另外用DSDCN (DSDCN-S) 替代标准卷积, 如图2 (b) 所示。实验表明, 性能从优到劣依次为DSDCN-S、Conv-S、扩张卷积-S、Conv-P和Conv。这说明我们的多尺度块嵌入能够灵活地嵌入图像块。如图8所示, 我们可视化了网络第一阶段特征图的平均值, 发现多尺度令牌 (DSDCN-S) 相比单尺度令牌 (Conv) 能帮助网络获得更丰富的信息。得益于深度可分离设计, 与深度可分离卷积相比, 我们的DSDCN在参数量和计算成本上仅有极小的增加。

多尺度注意力细化模块的有效性。表3表明, 我们的MSAR模块相比未使用MSAR模块的对照方案带来了1.97dB的有利增益, 而参数量 (0.1M) 和乘法运算次数 (0.905G) 仅略有增加。当我们将门控张量设为 $G \in \mathbb{R}^{h \times 1 \times H \times W} \rightarrow G \in \mathbb{R}^{h \times C \times H \times W}$ 时, 峰值信噪比和结构相似性仅获得微小提升, 却以参数量和计算复杂度的大幅增加为代价。综合考虑性能和计算负担, 我们设置 $G \in \mathbb{R}^{h \times 1 \times H \times W}$ 。

与其他线性自注意力模块的比较。表4将所提出的 T-MSA与几种常见的线性自注意力模块进行了比较。结果表明, TaylorFormer相比现有线性自注意力模块具有显著优势, 这归因于T-MSA能够建模远距离像素并逼近softmax注意力。

近似误差分析。为了探究近似误差及其影响, 我们研究了其效

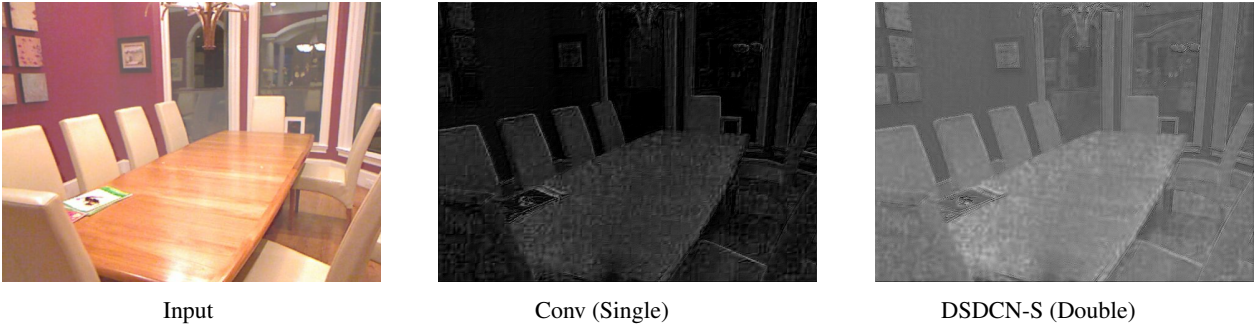


Figure 8: **Comparison of feature map visualization for single-scale and multi-scale tokens.** The design of multi-scale and multi-branch can capture more powerful features.

Table 6: **Results on the CSD dataset [8] for snow removal.** The best and second best results are highlighted in bold and underlined, respectively.

Methods	CSD (2000)		Overhead	
	PSNR↑	SSIM↑	#Param↓	MACs↓
DesnowNet [38]	20.13	0.81	26.15M	1.7KG
CycleGAN [17]	20.98	0.80	7.84M	42.4G
All in One [33]	26.31	0.87	44.00M	12.3G
HDCW-Net [8]	29.06	0.91	6.99M	9.8G
TKL [9]	33.89	0.96	31.35M	41.6G
SMGARN [10]	31.93	0.95	6.86M	450.3G
Uformer [73]	33.80	0.96	9.03M	19.8G
Restormer [78]	<u>35.43</u>	<u>0.97</u>	26.10M	141.0G
Ours (-B)	37.10	0.98	2.68M	38.5G

Table 7: **Results on the RainCityscapes dataset [25] for rain removal.** “-” indicates that the result is not available. The best and second best results are highlighted in bold and underlined, respectively.

	RESCAN [34]	DAFNet [25]	PCNet [27]	EPRRNet [83]	Uformer [73]	Restormer [78]	Ours(-B)
PSNR↑	28.97	30.06	24.49	31.11	32.26	<u>34.37</u>	36.55
SSIM↑	0.885	0.953	0.946	0.974	0.981	<u>0.988</u>	0.990
#Params↓	0.15M	-	0.63M	137.10M	9.03M	26.10M	2.68M
MACs↓	9.7G	-	2.4G	182.2G	19.8G	141.0G	38.5G

fect of different orders of Taylor expansion for softmax attention. Considering that the associative law is not applicable to the second-order T-MSA (T-MSA-2nd), which leads to a significant computational burden, we perform first-order and second-order Taylor expansions for Swin. Table 5 shows that T-MSA can effectively approximate softmax-attention, and the T-MSA-2nd is already very close to softmax-attention in performance. However, the computational complexity of both T-MSA-2nd and softmax-attention increases quadratically with image resolution, so it is difficult to model long-range pixel relationships in practical applications. Thus, we finally choose T-MSA-1st for our proposed method.

4.5. The Generalization Capabilities

Though this paper focuses on image dehazing, we have also evaluated the generalization capabilities of our pro-

posed model on other tasks, specifically snow and rain removal, using the CSD [8] and RainCityscapes [25] datasets. The setting of the training and testing dataset for snow removal follows the existing methods [8, 10], and the dataset setting for rain removal follows existing methods [25, 83]. The results, presented in Table 6 and Table 7, demonstrate that our model, MB-TaylorFormer, performs well in these tasks, indicating that its capabilities are not limited to dehazing. The supplementary material includes visualization results for these tasks.

5. Conclusion

In this paper, we propose a multi-branch linearized Transformer network, called MB-TaylorFormer, which consists of multi-scale patch embedding and Taylor expansion of self-attention. Multi-scale patch embedding with flexible shape of receptive field, multi-scale size receptive field, and multi-level semantic information, can enable flexible embedding of diverse visual tokens. The Taylor expansion of softmax-attention using the matrix multiplicative associate law reduces the computational complexity. Further, we correct the output of self-attention by gating attention, which allows MB-TaylorFormer to perform both long-range attention and local corrections. Experimental results on various datasets demonstrate the effectiveness, lightness and generalization of the proposed MB-TaylorFormer.

6. Acknowledgements

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图8：单尺度与多尺度令牌的特征图可视化对比。多尺度和多分支的设计能够捕获更强大的特征。

表6：在CSD数据集上的结果 [8]，用于除雪任务。最佳和次佳结果分别以粗体和下划线标出。

方法	CSD (2000)		Overhead	
	峰值信噪比↑	结构相似性↑	#参数量↓	乘法运算次数↓
DesnowNet [38]	20.13	0.81	26.15M	1.7KG
CycleGAN [17]	20.98	0.80	7.84M	42.4G
All in One [33]	26.31	0.87	44.00M	12.3G
HDCW-Net [8]	29.06	0.91	6.99M	9.8G
TKL [9]	33.89	0.96	31.35M	41.6G
SMGARN [10]	31.93	0.95	6.86M	450.3G
Uformer [73]	33.80	0.96	9.03M	19.8G
Restormer [78]	<u>35.43</u>	<u>0.97</u>	26.10M	141.0G
本文方法 (-B)	37.10	0.98	2.68M	38.5G

表 7：RainCityscapes 数据集上的除雨结果 [25]。“-”表示结果不可用。最佳和次佳结果分别以粗体和下划线标出。

	RESCAN [34]	DAFNet [25]	PCNet [27]	EPRRNet [83]	Uformer [73]	Restormer [78]	本文方法 (-B)
峰值信噪比↑	28.97	30.06	24.49	31.11	32.26	<u>34.37</u>	36.55
结构相似性↑	0.885	0.953	0.946	0.974	0.981	<u>0.988</u>	0.990
参数量↓	0.15M	-	0.63M	137.10M	9.03M	26.10M	2.68M
乘法运算次数↓	9.7G	-	2.4G	182.2G	19.8G	141.0G	38.5G

泰勒展开的不同阶数对 softmax 注意力的影响。考虑到结合律不适用于二阶 T-MSA (T-MSA-2nd)，这会导致显著的计算负担，我们对 Swin 进行一阶和二阶泰勒展开。表 5 显示，T-MSA 能够有效近似 softmax 注意力，并且 T-MSA-2nd 在性能上已经非常接近 softmax 注意力。然而，T-MSA-2nd 和 softmax 注意力的计算复杂度都随图像分辨率呈二次方增长，因此在实际应用中很难建模长距离像素关系。因此，我们最终选择 T-MSA-1st 作为所提出方法的方案。

4.5. 泛化能力

尽管本文主要关注图像去雾，但我们也评估了所提模型在其他任务上的泛化能力，

具体包括使用CSD [8]和RainCityscapes [25]数据集进行除雪和除雨任务。除雪任务的训练和测试数据集设置遵循现有方法 [8, 10]，除雨任务的数据集设置遵循现有方法 [25, 83]。表6和表7中呈现的结果表明，我们的模型MB-TaylorFormer在这些任务中表现良好，说明其能力并不仅限于去雾。补充材料中包含了这些任务的可视化结果。

5. 结论

在本文中，我们提出了一种多分支线性化 Transformer网络，称为MB-TaylorFormer，它由多尺度块嵌入和自注意力的泰勒展开组成。具有灵活感受野形状、多尺度感受野大小和多层次语义信息的多尺度块嵌入，能够实现多样视觉令牌的灵活嵌入。利用矩阵乘法结合律对softmax注意力进行泰勒展开，降低了计算复杂度。此外，我们通过门控注意力校正自注意力的输出，使MB-TaylorFormer能够同时执行长程注意力和局部校正。在多个数据集上的实验结果表明，所提出的MB-TaylorFormer具有有效性、轻量性和泛化能力。

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